

Customer Segmentation and Review Score Prediction in Olist E-Commerce

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Problem

Olist faces the challenge of sustaining customer satisfaction and loyalty in a competitive e-commerce market, where delivery delays and inconsistent service quality often drive churn. This project explores how customer segmentation and review score prediction can guide strategies to improve retention, satisfaction, and lifetime value.

Solution

We aim to segment customers using RFM analysis and K-Means clustering to identify loyal, at-risk, and one-time buyers, and build predictive models of review scores to uncover key drivers of satisfaction. These insights enable Olist to prioritize delivery reliability, design targeted retention programs, and optimize payment offerings to boost loyalty and lifetime value.

Dataset

We are using the Olist Brazilian E-Commerce Public Dataset transactional data from 100k+ orders across Brazil (2016–2018), covering customers, orders, products, payments, reviews, and sellers.

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Data Wrangling

Data Wrangling

- Converted all timestamp fields (purchase, approval, shipping, delivery, review) into datetime objects for time-based analysis.
- Handled missing values: removed critical nulls (e.g., delivery dates, review scores) and retained non-critical ones.
- Replaced Portuguese product categories with English translations for interpretability.
- Removed duplicates and implausible records (e.g., negative delivery times, zero-priced orders).
- Integrated 8 relational tables (customers, orders, items, payments, reviews, products, sellers, geolocation) into a unified dataset.

Exploring the Data

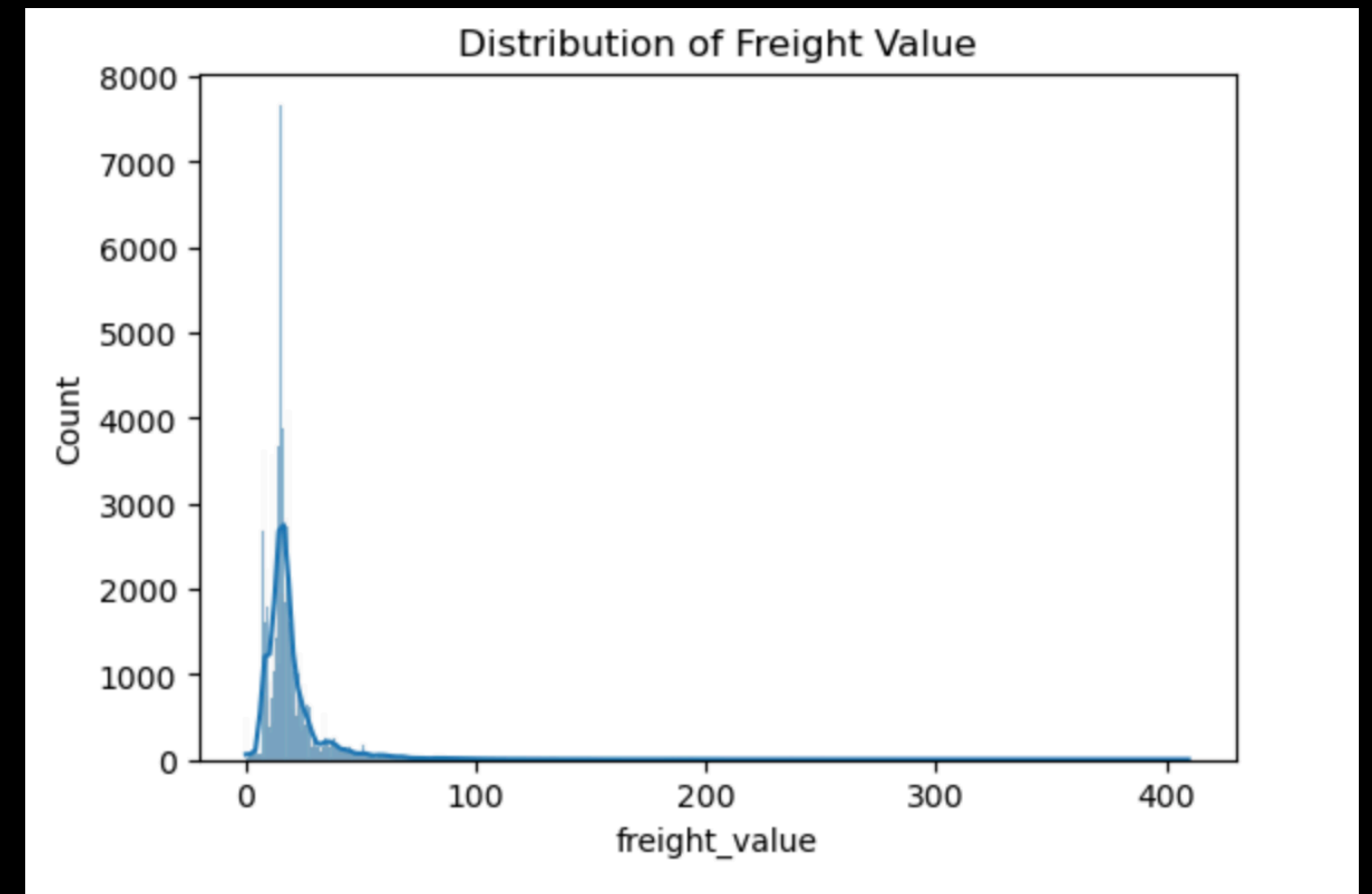
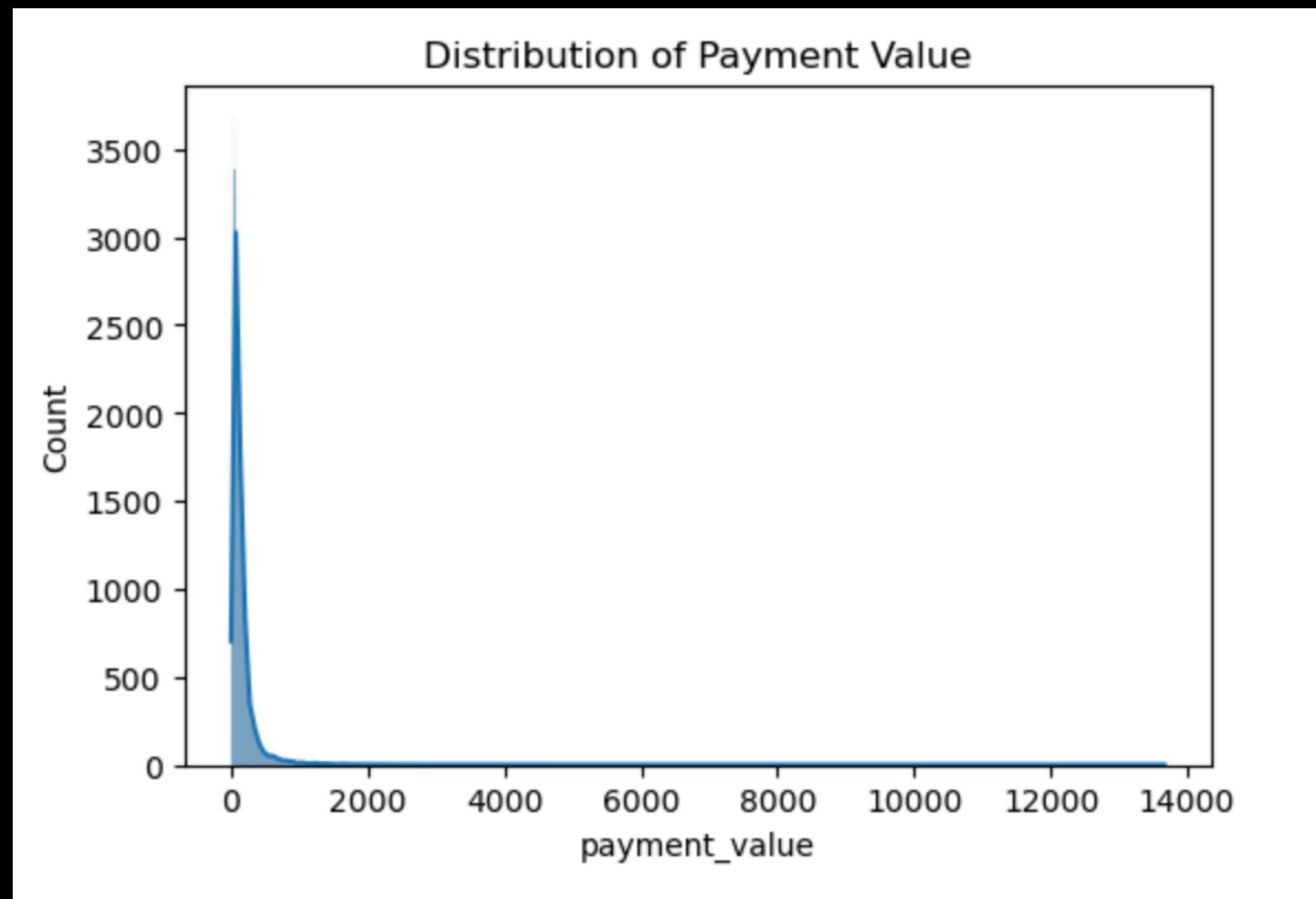
Exploring the Data

Univariate Analysis



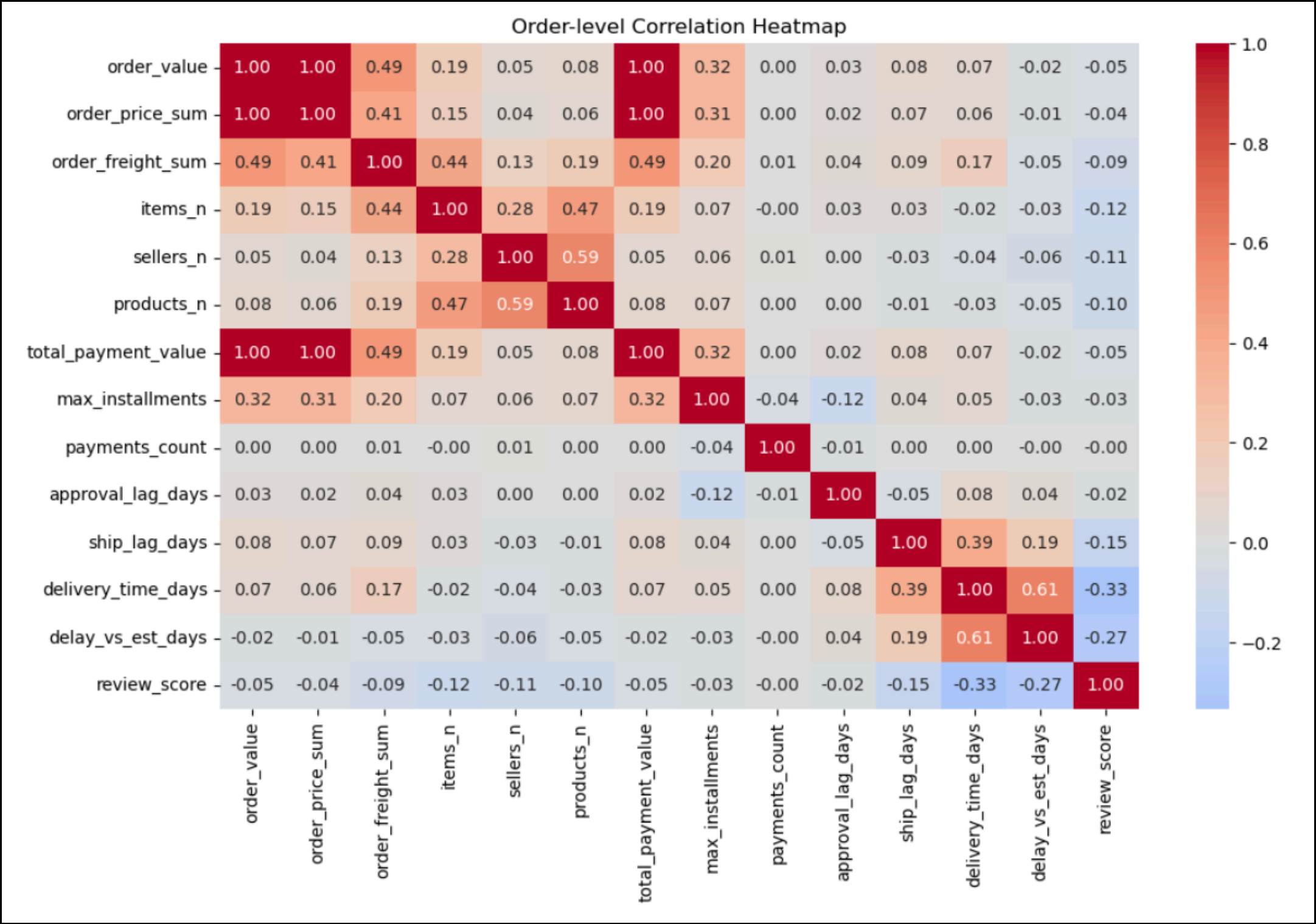
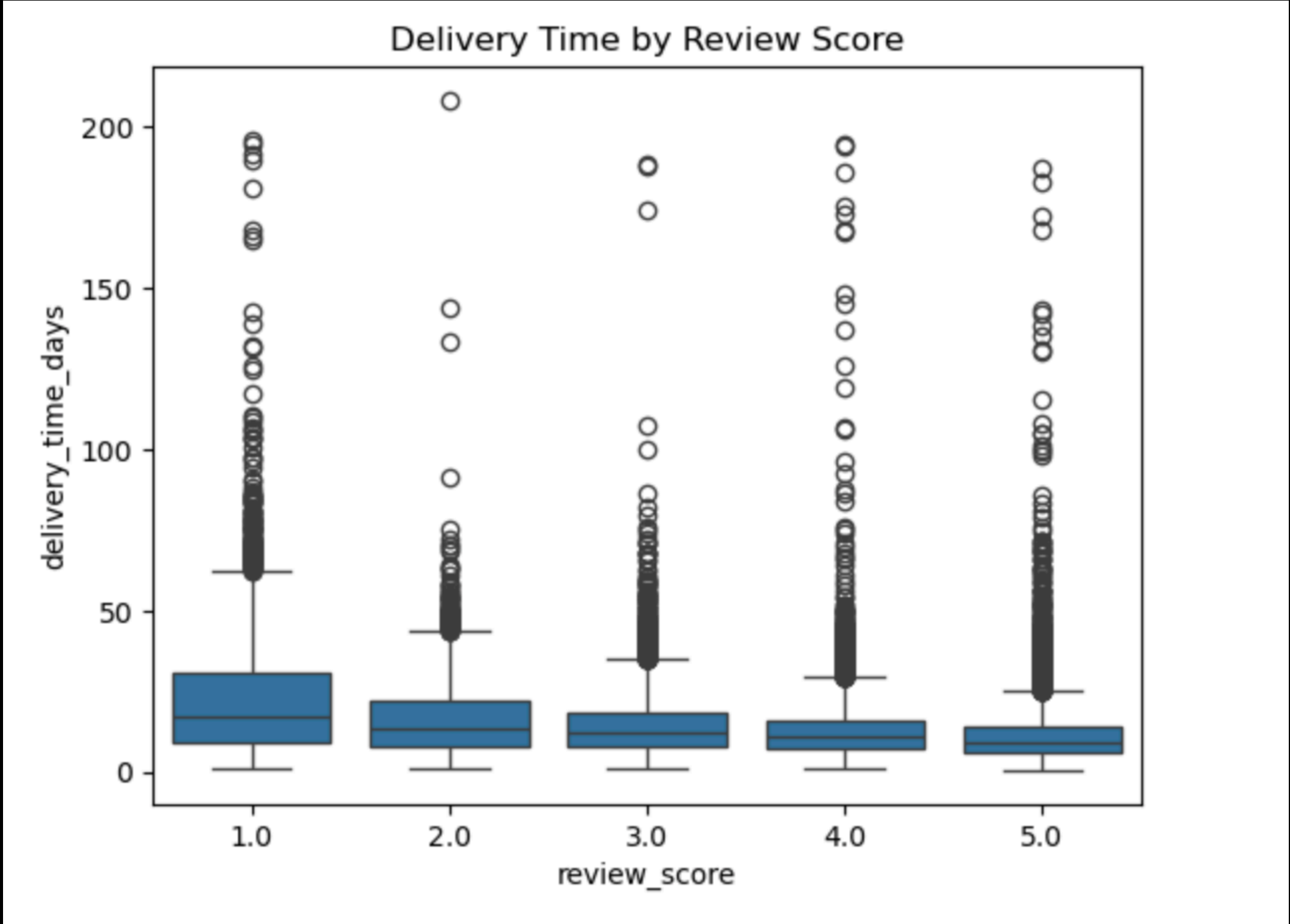
Exploring the Data

Univariate Analysis



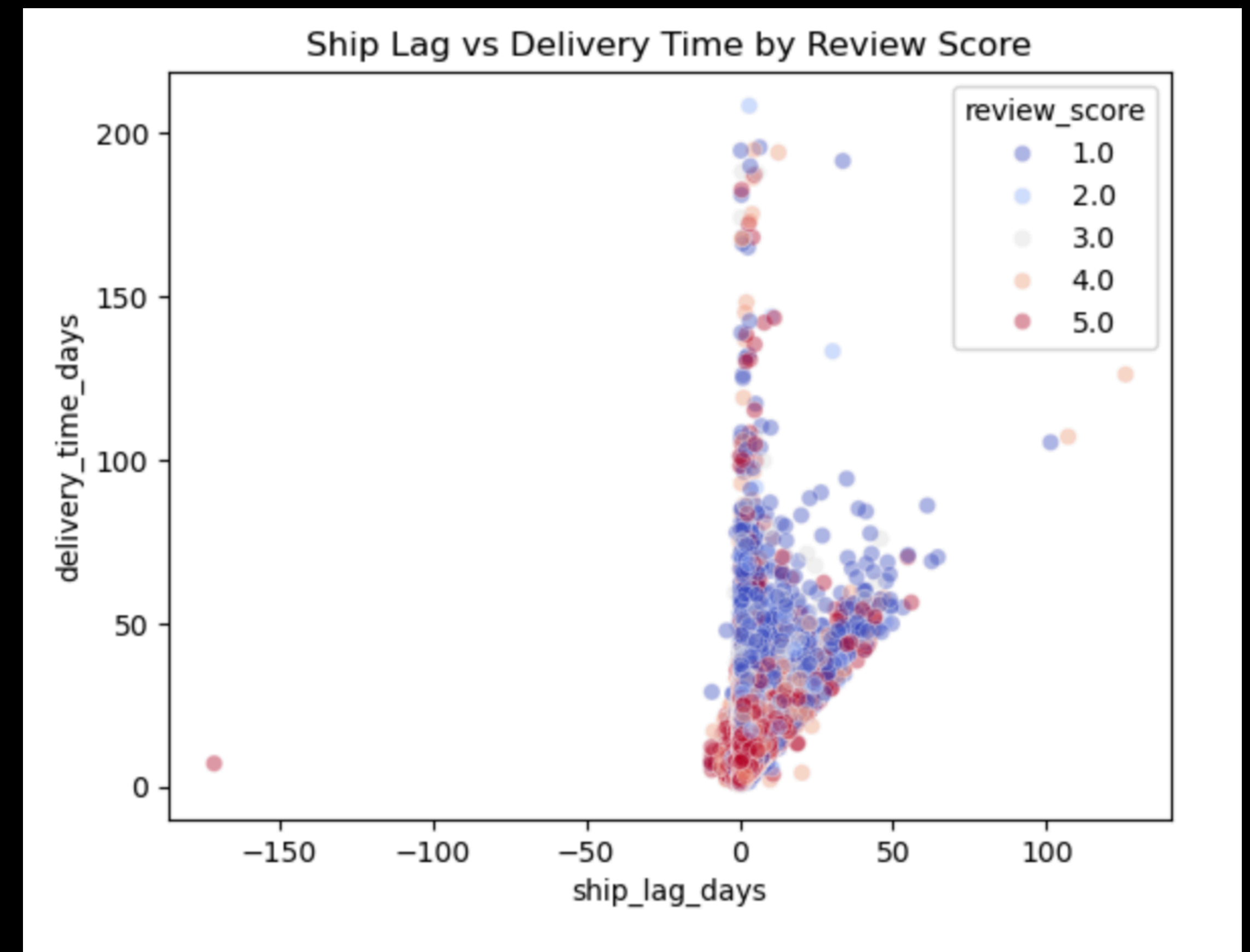
Exploring the Data

Review Scores and Satisfaction



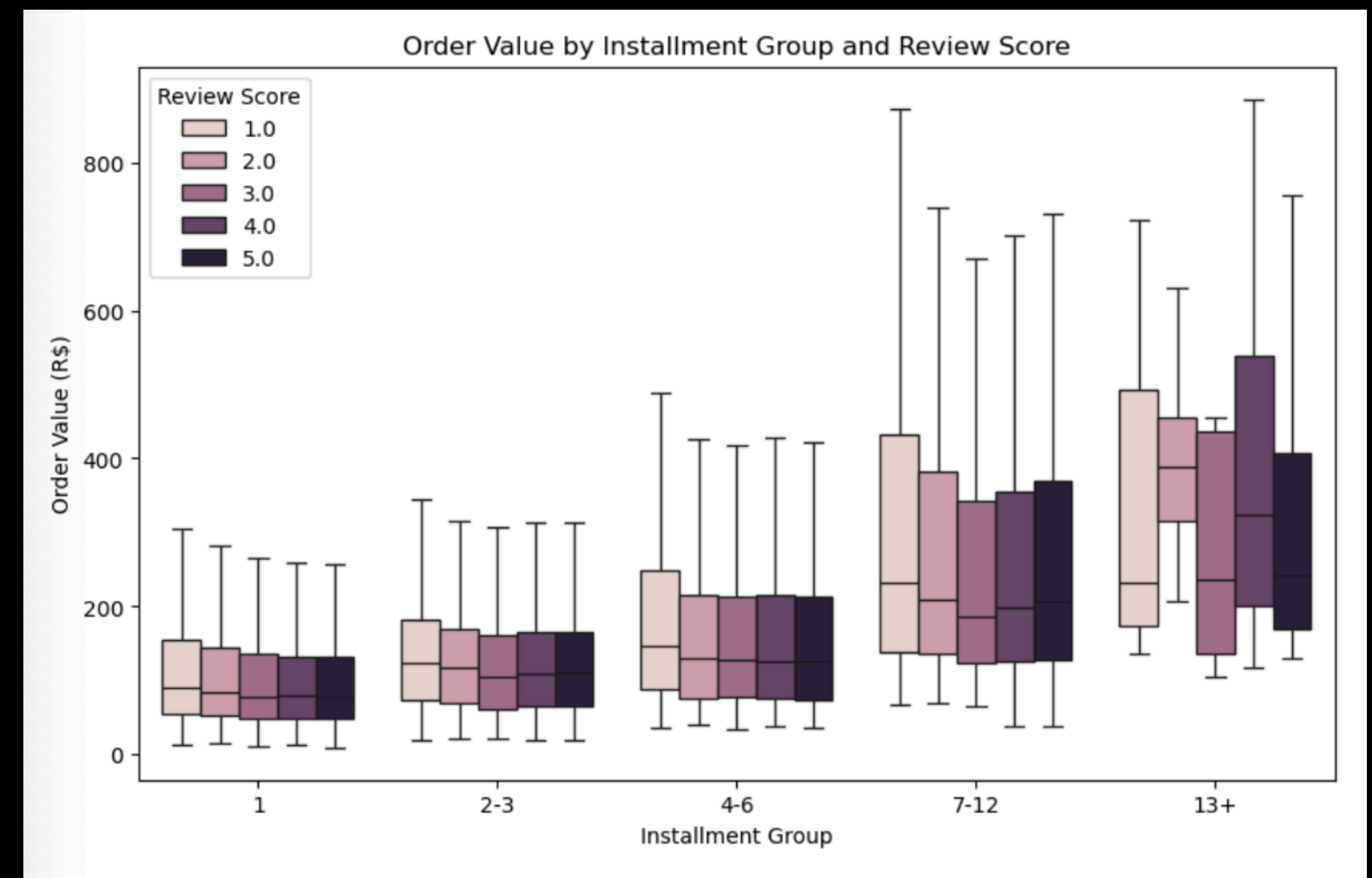
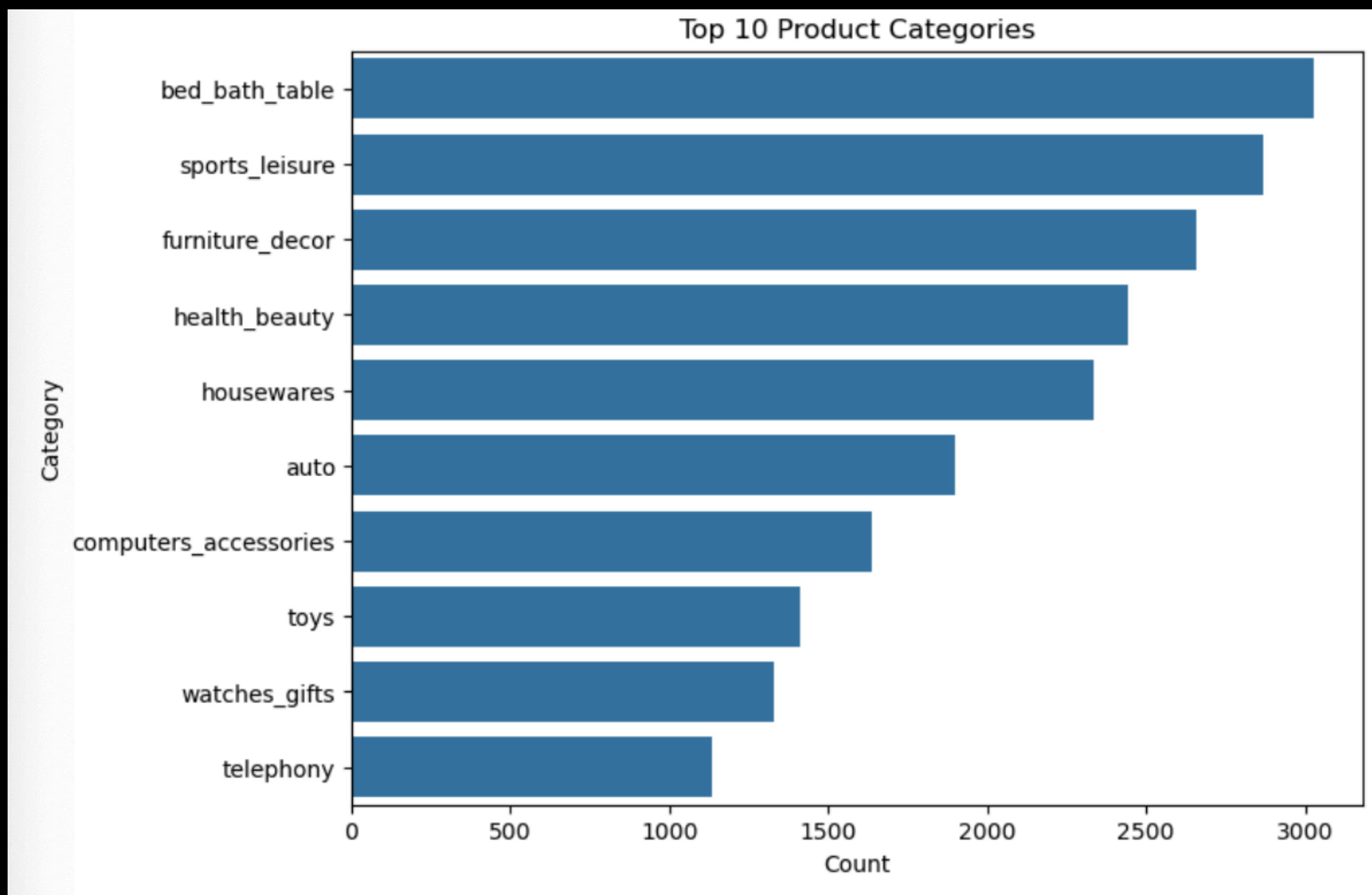
Exploring the Data

Delivery Performance



Exploring the Data

Order Characteristics



Feature Engineering & Preprocessing

Feature Engineering & Proprocessing

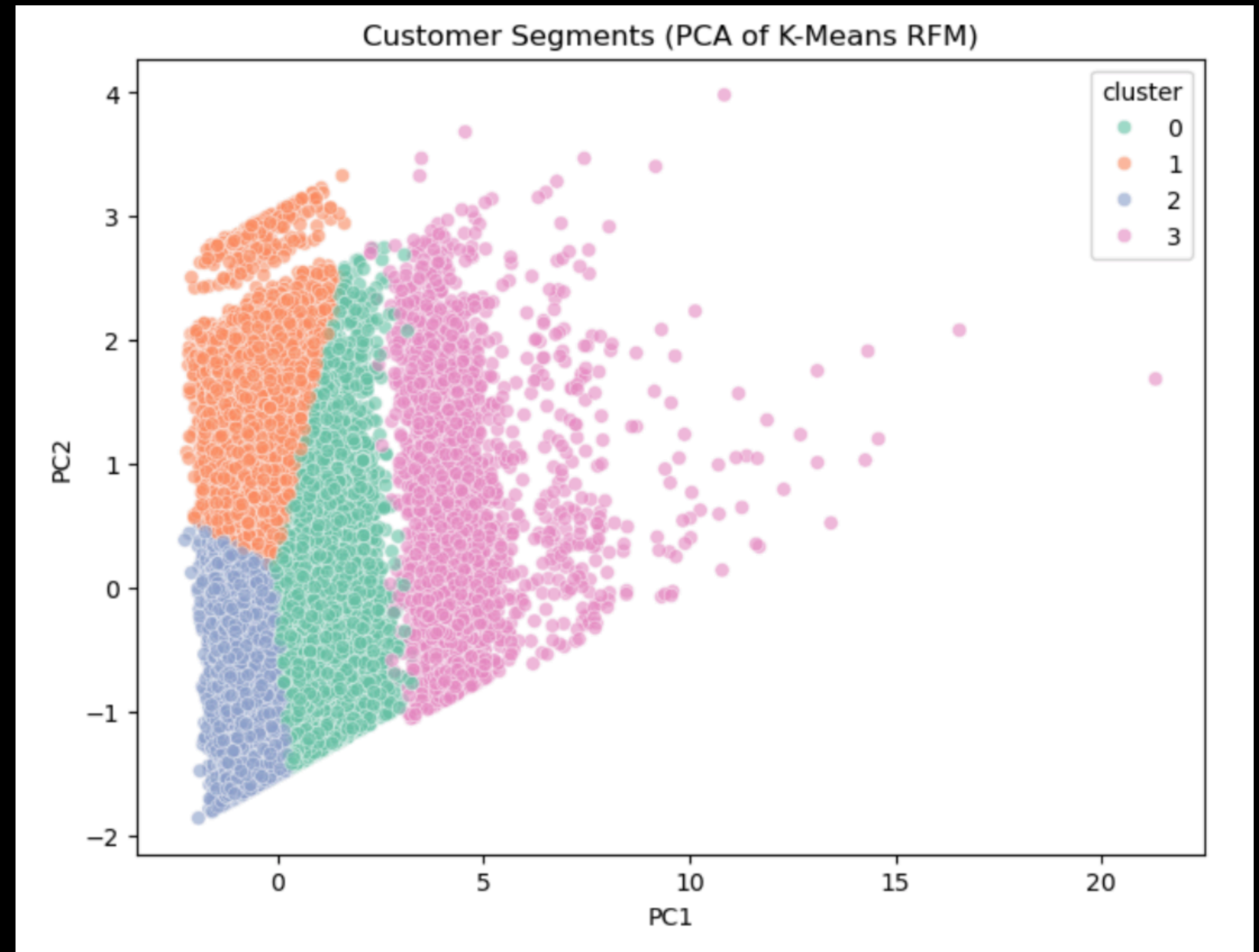
- Created RFM metrics (Recency, Frequency, Monetary) to capture customer behavior.
- Engineered delivery delay (actual vs. estimated delivery date) as a key operational feature.
- Calculated order value as product price + freight value.
- Encoded payment methods into dummy variables (credit card, boleto, debit, voucher).
- Standardized continuous features (RFM, order value, delivery delay) for clustering and modeling.
- One-hot encoded categorical variables for use in machine learning models.
- Split dataset into training and test sets for model evaluation.

Modeling

Modeling

Customer Segmentation (Clustering)

- Applied K-Means clustering on standardized RFM features (Recency, Frequency, Monetary).
- Used elbow method and silhouette scores to select 4-cluster solution.
- Segments: Champions, Loyal, At-Risk, Lost.
- Revealed that a small minority of high-value customers drive most revenue.

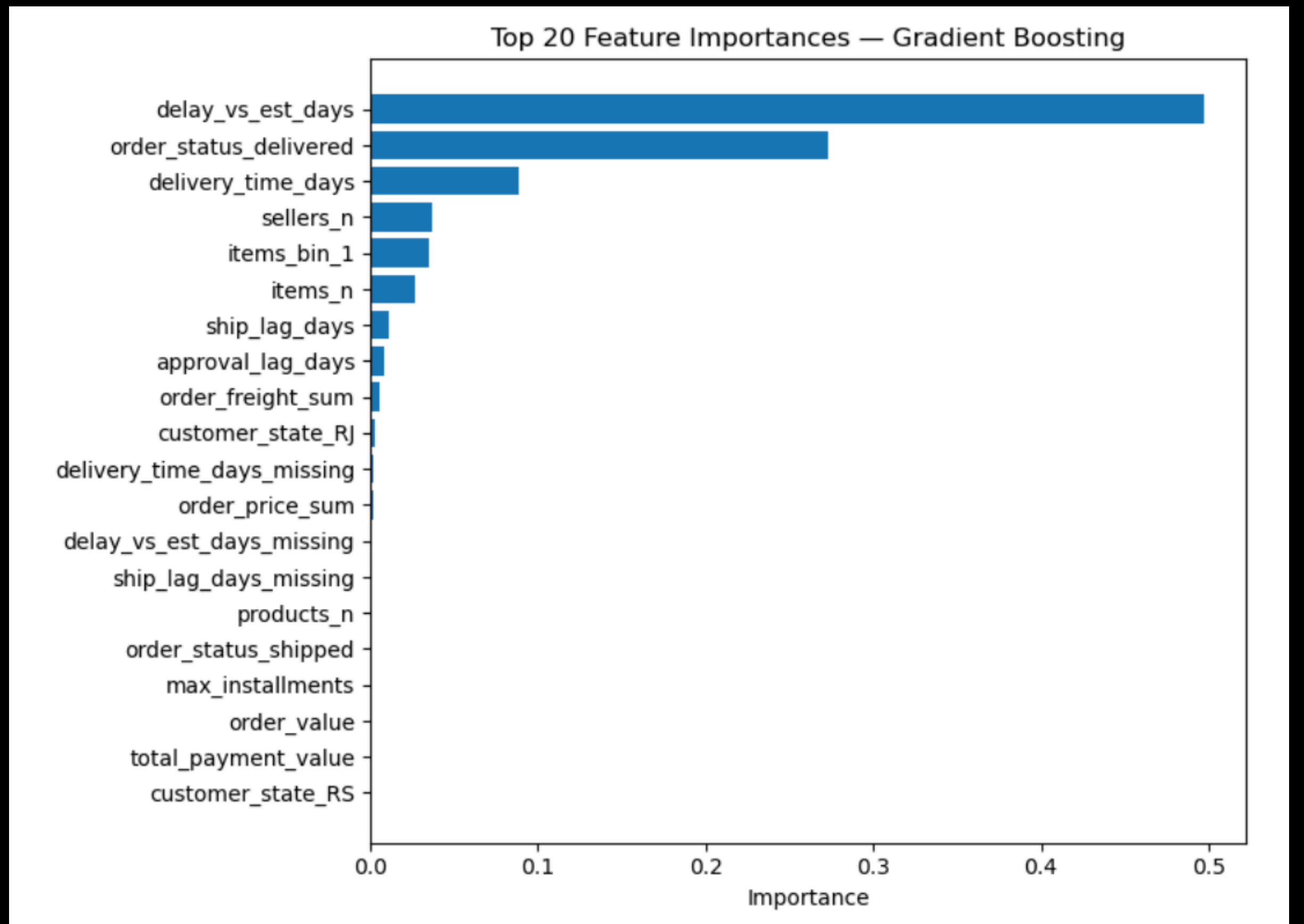


Modeling

Satisfaction Prediction (Regression)

- Tested Linear Regression, Random Forest, Gradient Boosting models.
- Evaluated using R^2 , RMSE, MAE on test set.
- Gradient Boosting performed best ($R^2 = 0.26$).
- Feature importance: Delivery delays strongest predictor.

Model	RMSE	MAE	R^2
Baseline (Mean)	1.324	1.035	0.000
Linear Regression	1.184	0.911	0.200
Random Forest	1.159	0.893	0.234
Gradient Boosting	1.139	0.873	0.260



Conclusions

- Majority of customers are one-time buyers with low long-term value.
- Small group of loyal, high-spending customers drives most revenue.
- Delivery performance is the strongest driver of satisfaction.
- Targeted retention programs can reduce churn and boost lifetime value.

Next Steps

- Invest in delivery reliability and transparency.
- Develop loyalty and win-back programs for high-value and at-risk customers.
- Explore alternative clustering methods (DBSCAN, hierarchical) for deeper insights.
- Incorporate additional data sources (demographics, marketing exposure, product quality) to improve models.

Q & A